

Report Trial LC 34 In Plating Line Of Niterra Indonesia Overview

Updated 14-10-2025

Prepared by:

Production Engineer

- 1. Background**
- 2. LC-34 Degreasing Performance (Trial Masspro in Line)**
- 3. WWT Performance – Lab Scale (Lab Check vs Internal Check)**
- 4. WWT Performance – Trial Masspro Scale (Lab Check vs Internal Check)**

1. Background, CT 15046

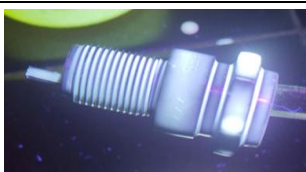
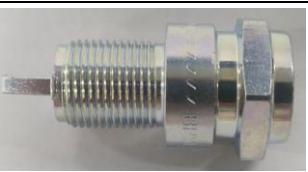
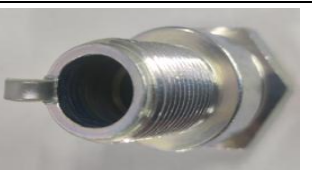
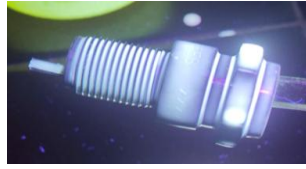
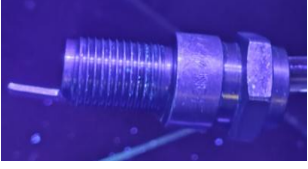
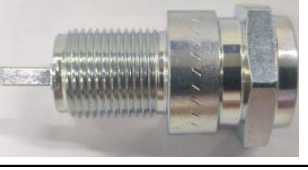
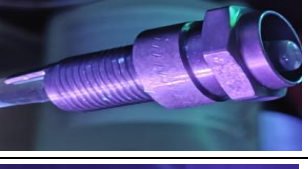
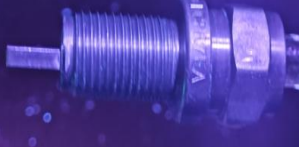
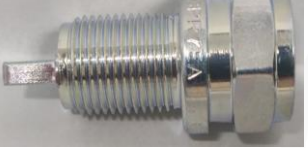
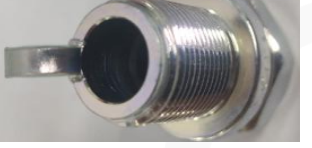
配付先 Destination	連絡のみ For your information	技術連絡書 Engineering Communication Sheet	№	CT- 15046	P.1/2
	要回答 Response requested		2025 年 7 月 2 日 July 2, 2025		
(03)	受信部署 To	発信部署 From	承認 Approved	点検 Checked	作成 Prepared
(11)	関連部署 殿 Relevant department/plant	IGNITE BU 第2技術部 金具中軸設計課 Process Engineering Dept., IGNITE Business Unit		—	
(14)	件名 Subject	インドネシア脱脂液代替ラインテスト Indonesia Alternative Degreasing Solution Line Test			
(30)	要素開発コード Element development code	—	初品重要度ランク Level of importance for initial products	—	
連絡事項 / Messages ・インドネシアめっきラインでの代替脱脂液ラインテスト方法に関してご連絡いたします。 ・We would like to inform you about the line test method for the alternative degreasing solution on the plating line in Indonesia. ・テスト実施に当たり各項目の詳細内容は添付のP1～P12を参照ください。 ・For the details of each item related to the test implementation, please refer to the attached pages P1 to P12.					
対象 / Flow method 工場: Niterra-インドネシア/Factory: Niterra Indonesia 工程: めっき工程/Process: Plating Process					
事前準備、事前確認/Pre-Test Preparation, Pre-Test Check (事前準備/Pre-Test Preparation) テストに必要な準備物等に関して、添付P2～P4を参照 Please refer to the attached pages P2 to P4 for the necessary preparations and items for the test.					
事前確認/Pre-Test Check ○脱脂液分析方法/分析方法妥当性確認 ・LC-34の分析方法の確認と分析精度妥当性確認を添付P5、P6を参考に確認ください。 ・Please confirm the LC-34 analysis method and the validity of its analytical accuracy with reference to the attached pages P5 and P6.					
○廃水処理 (WWT) への影響の確認 ○Confirmation of impact on wastewater treatment (WWT) ・脱脂液をLC-34に切替える当たって排水性の確認を添付P.7を参考に確認ください。 ・Please refer to the attached page P.7 to confirm the wastewater compatibility when switching the degreasing solution to LC-34.					
○蛍光探傷液の塗布方法/Fluorescent Penetrant Application Method ・ラインテスト実施前に、添付10を参考に転造後金具に蛍光探傷液入りの転造油を塗布ください。 ・Before conducting the line test, please apply rolling oil containing fluorescent penetrant to the rolled metal fittings, referring to Attachment 10.					

技術連絡書 Engineering Communication Sheet			№	CT- 15046	P.2/2
ラインテスト/Line Test ・テストフロー及び添付P.8～P.12に基づき、No1～No5のテストを実施ください。 ・Based on the test flow and Attachments P.8 through P.12, please perform Tests No. 1 to No. 5. ※Test No1～4は、浸漬脱脂後と電解脱脂後の最終水洗で金具を採取しブラックライトによる脱脂性確認と写真撮りを実施のこと。詳細は添付P9、P10参照のこと。 ※For Tests No. 1 to 4, after immersion and electrolytic degreasing, collect the metal fittings following the final rinse. Perform degreasing performance verification using a black light and take photographs. Please refer to the attached pages P9 and P10 for further details.					
Test No	確認項目 Check Items	操作 operation	※浸漬脱脂 (immersion)	※電解脱脂 (electrolysis)	
1	脱脂性確認(既存) Degreasing Effectiveness Check	手動 manual	脱脂39/Degreasing 39 濃度/Concentration: 40g/L 温度/Temperature: 60℃	PN36/PN-36 濃度/Concentration: 40g/L 温度/Temperature: 60℃	
2	脱脂性確認(LC-34) Degreasing Effectiveness Check	手動 manual	LC-34 濃度/Concentration: 79g/L 温度/Temperature: 49℃	LC-34 濃度/Concentration: 39g/L 温度/Temperature: 49℃	
3	脱脂性確認(LC-34) Degreasing Effectiveness Check	手動 manual	LC-34 濃度/Concentration: 69g/L 温度/Temperature: 49℃	LC-34 濃度/Concentration: 39g/L 温度/Temperature: 49℃	
4	脱脂性確認(LC-34) Degreasing Effectiveness Check	手動 manual	LC-34 濃度/Concentration: 59g/L 温度/Temperature: 49℃	LC-34 濃度/Concentration: 39g/L 温度/Temperature: 49℃	
5	密着性確認 Adhesion Check	自動 automatic	LC-34 濃度/Concentration: 69(79)g/L 温度/Temperature: 49℃	LC-34 濃度/Concentration: 39g/L 温度/Temperature: 49℃	
テスト評価と記録/Line Test Evaluation and Record ・ラインテスト時の評価と記録に関しては添付P11を参照ください。 (テスト時の脱脂液濃度、温度の確認、脱脂性確認の写真、外観と密着性確認等) ・Please refer to Attachment P11 for the evaluation and recording during the line test. (Confirmation of degreasing solution concentration and temperature during the test, photos of degreasing performance verification, and inspection of appearance and adhesion.)					
テストフロー/Test Flow ※添付P.12参照					

Background:

Our current degreasing chemical (Dasshi 39 & PN 36) will be discontinued in the future. Based on CT 15046 we must do several trial of LC-34 degreasing as replacement.

2. LC-34 Degreasing Performance, Black Light & Adhesion Test

No.	Degreasing Process		E-Degreasing Process		Evaluation (Visual)					Judgement
	Conc. (gr/L)	Temp. (°C)	Conc. (gr/L)	Temp. (°C)	UV - Test			After Plating	Adhesion Test (PSB-035 Item no. 10)	
					Before	After Degreasing	After E-Degreasing			
1.	40,8	60,5	40,8	60,3						OK
2.	78,78	50,5	39,96	50						OK
3.	69,65	49,8	39,96	50,5						OK
4.	58,23	50	39,96	49,5						OK (There is small dots of fluorescent)
5.	78,78	50,5	39,96	50						OK

Conclusion:

- Black light check show that fluorescent can almost completely cleaned after E-degreasing in every concentration & plating products resulted in satisfactory output
- In Matrix no 4 (59 gr/L setup) we consider this OK due to during Oil & Fluorescent mixing the Fluorescent volume is 10 times higher (1L), if we use the mentioned amount in CT-15406 (100 mL) it is very likely there will be no fluorescent residue

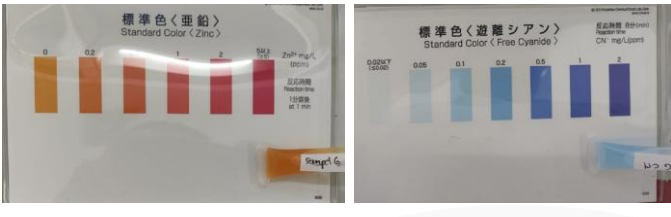
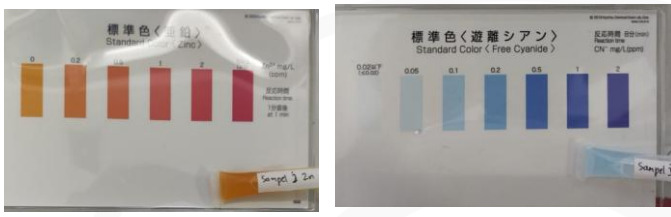
2. LC-34 Degreasing Performance, N30 Result

Matrix 1	Matrix 2	Matrix 3	Matrix 4	Matrix 5
INTEGRASI KUSI PERTAMA / PRODUCTION No Dsk: PK-GA-GA-30 Revisi: 00 Tanggal: 05 APRIL 2023 Halaman: 1/2 Part: CHUCKER MISHILL NAME: METAL SHELL OTHERS: PLUS CAP ASSY TANGGAL DATA: Monday, 23 July 2025 NAMA PROSES: PLATING TYPE/ PART NO.: SPSES QUANTITY/LOT SIZE: 30 PCS SAMPLE DATA: Monday, 23 July 2025 TANGGAL PROSES: Monday, 23 July 2025 NO. LOT PRODUKSI: - DEFECT RATE: - Part: NO BARU MESIN BARU PERUBAHAN PROSES PERUBAHAN DESAIN PERUBAHAN MATERIAL INITIAL MASS PRODUCTION LAIN-LAIN DETAIL PROPOSAL TIBAL LC 34 Matrix 1	INTEGRASI KUSI PERTAMA / PRODUCTION No Dsk: PK-GA-GA-30 Revisi: 00 Tanggal: 05 APRIL 2023 Halaman: 1/2 Part: CHUCKER MISHILL NAME: METAL SHELL OTHERS: PLUS CAP ASSY TANGGAL DATA: Monday, 23 July 2025 NAMA PROSES: PLATING TYPE/ PART NO.: SPSES QUANTITY/LOT SIZE: 30 PCS SAMPLE DATA: Monday, 23 July 2025 TANGGAL PROSES: Monday, 23 July 2025 NO. LOT PRODUKSI: - DEFECT RATE: - Part: NO BARU MESIN BARU PERUBAHAN PROSES PERUBAHAN DESAIN PERUBAHAN MATERIAL INITIAL MASS PRODUCTION LAIN-LAIN DETAIL PROPOSAL TIBAL LC 34 Matrix 2	INTEGRASI KUSI PERTAMA / PRODUCTION No Dsk: PK-GA-GA-30 Revisi: 00 Tanggal: 05 APRIL 2023 Halaman: 1/2 Part: CHUCKER MISHILL NAME: METAL SHELL OTHERS: PLUS CAP ASSY TANGGAL DATA: Monday, 23 July 2025 NAMA PROSES: PLATING TYPE/ PART NO.: SPSES QUANTITY/LOT SIZE: 30 PCS SAMPLE DATA: Monday, 23 July 2025 TANGGAL PROSES: Monday, 23 July 2025 NO. LOT PRODUKSI: - DEFECT RATE: - Part: NO BARU MESIN BARU PERUBAHAN PROSES PERUBAHAN DESAIN PERUBAHAN MATERIAL INITIAL MASS PRODUCTION LAIN-LAIN DETAIL PROPOSAL TIBAL LC 34 Matrix 3	INTEGRASI KUSI PERTAMA / PRODUCTION No Dsk: PK-GA-GA-30 Revisi: 00 Tanggal: 05 APRIL 2023 Halaman: 1/2 Part: CHUCKER MISHILL NAME: METAL SHELL OTHERS: PLUS CAP ASSY TANGGAL DATA: Monday, 23 July 2025 NAMA PROSES: PLATING TYPE/ PART NO.: SPSES QUANTITY/LOT SIZE: 30 PCS SAMPLE DATA: Monday, 23 July 2025 TANGGAL PROSES: Monday, 23 July 2025 NO. LOT PRODUKSI: - DEFECT RATE: - Part: NO BARU MESIN BARU PERUBAHAN PROSES PERUBAHAN DESAIN PERUBAHAN MATERIAL INITIAL MASS PRODUCTION LAIN-LAIN DETAIL PROPOSAL TIBAL LC 34 Matrix 4	INTEGRASI KUSI PERTAMA / PRODUCTION No Dsk: PK-GA-GA-30 Revisi: 00 Tanggal: 05 APRIL 2023 Halaman: 1/2 Part: CHUCKER MISHILL NAME: METAL SHELL OTHERS: PLUS CAP ASSY TANGGAL DATA: Monday, 23 July 2025 NAMA PROSES: PLATING TYPE/ PART NO.: SPSES QUANTITY/LOT SIZE: 30 PCS SAMPLE DATA: Monday, 23 July 2025 TANGGAL PROSES: Monday, 23 July 2025 NO. LOT PRODUKSI: - DEFECT RATE: - Part: NO BARU MESIN BARU PERUBAHAN PROSES PERUBAHAN DESAIN PERUBAHAN MATERIAL INITIAL MASS PRODUCTION LAIN-LAIN DETAIL PROPOSAL TIBAL LC 34 Matrix 5
ITEM: Plating Thickness ALAT: X-Ray STD: Min 5.00, Max 5.00, MSK X L25, No Defect 1: 5.130 OK OK OK OK 2: 4.580 OK OK OK OK 3: 5.190 OK OK OK OK 4: 4.830 OK OK OK OK 5: 5.190 OK OK OK OK 6: 5.130 OK OK OK OK 7: 5.180 OK OK OK OK 8: 4.880 OK OK OK OK 9: 4.880 OK OK OK OK 10: 5.280 OK OK OK OK 11: 5.630 OK OK OK OK 12: 5.480 OK OK OK OK 13: 4.250 OK OK OK OK 14: 4.880 OK OK OK OK 15: 4.980 OK OK OK OK 16: 5.080 OK OK OK OK 17: 5.630 OK OK OK OK 18: 4.300 OK OK OK OK 19: 4.880 OK OK OK OK 20: 5.480 OK OK OK OK 21: 4.380 OK OK OK OK 22: 5.140 OK OK OK OK 23: 5.380 OK OK OK OK 24: 5.080 OK OK OK OK 25: 5.480 OK OK OK OK 26: 5.430 OK OK OK OK 27: 4.880 OK OK OK OK 28: 5.430 OK OK OK OK 29: 5.380 OK OK OK OK 30: 5.180 OK OK OK OK MAX: 5.630 MIN: 4.250 X bar: 5.131 S bar: 0.130 Pp: 2.46 Ppk: 1.73 OK / NG: 6000 / 6000 APPROVED: ISPRIANTI, CHANDRA	ITEM: Plating Thickness ALAT: X-Ray STD: Min 5.00, Max 5.00, MSK X L25, No Defect 1: 5.050 OK OK OK OK 2: 5.040 OK OK OK OK 3: 5.100 OK OK OK OK 4: 4.830 OK OK OK OK 5: 5.100 OK OK OK OK 6: 4.740 OK OK OK OK 7: 4.880 OK OK OK OK 8: 5.130 OK OK OK OK 9: 5.430 OK OK OK OK 10: 4.880 OK OK OK OK 11: 5.430 OK OK OK OK 12: 5.080 OK OK OK OK 13: 5.030 OK OK OK OK 14: 5.370 OK OK OK OK 15: 4.880 OK OK OK OK 16: 4.880 OK OK OK OK 17: 5.280 OK OK OK OK 18: 5.140 OK OK OK OK 19: 5.230 OK OK OK OK 20: 5.480 OK OK OK OK 21: 4.880 OK OK OK OK 22: 4.930 OK OK OK OK 23: 4.780 OK OK OK OK 24: 4.990 OK OK OK OK 25: 5.480 OK OK OK OK 26: 5.370 OK OK OK OK 27: 5.130 OK OK OK OK 28: 5.080 OK OK OK OK 29: 5.030 OK OK OK OK 30: 4.930 OK OK OK OK MAX: 5.480 MIN: 4.780 X bar: 5.09 S bar: 0.13 Pp: 2.37 Ppk: 1.81 OK / NG: 6000 / 6000 APPROVED: ISPRIANTI, CHANDRA	ITEM: Plating Thickness ALAT: X-Ray STD: Min 5.00, Max 5.00, MSK X L25, No Defect 1: 5.140 OK OK OK OK 2: 5.030 OK OK OK OK 3: 5.100 OK OK OK OK 4: 4.830 OK OK OK OK 5: 5.080 OK OK OK OK 6: 4.880 OK OK OK OK 7: 4.880 OK OK OK OK 8: 4.830 OK OK OK OK 9: 5.130 OK OK OK OK 10: 5.080 OK OK OK OK 11: 5.430 OK OK OK OK 12: 4.980 OK OK OK OK 13: 4.980 OK OK OK OK 14: 4.780 OK OK OK OK 15: 5.080 OK OK OK OK 16: 4.820 OK OK OK OK 17: 4.780 OK OK OK OK 18: 5.430 OK OK OK OK 19: 5.080 OK OK OK OK 20: 4.780 OK OK OK OK 21: 5.180 OK OK OK OK 22: 5.030 OK OK OK OK 23: 5.080 OK OK OK OK 24: 5.080 OK OK OK OK 25: 4.880 OK OK OK OK 26: 5.130 OK OK OK OK 27: 4.930 OK OK OK OK 28: 5.480 OK OK OK OK 29: 5.180 OK OK OK OK 30: 5.170 OK OK OK OK MAX: 5.480 MIN: 4.780 X bar: 5.09 S bar: 0.13 Pp: 2.43 Ppk: 1.79 OK / NG: 6000 / 6000 APPROVED: ISPRIANTI, CHANDRA	ITEM: Plating Thickness ALAT: X-Ray STD: Min 5.00, Max 5.00, MSK X L25, No Defect 1: 5.140 OK OK OK OK 2: 5.030 OK OK OK OK 3: 5.100 OK OK OK OK 4: 4.830 OK OK OK OK 5: 5.080 OK OK OK OK 6: 4.880 OK OK OK OK 7: 4.880 OK OK OK OK 8: 4.830 OK OK OK OK 9: 5.130 OK OK OK OK 10: 5.080 OK OK OK OK 11: 5.430 OK OK OK OK 12: 4.980 OK OK OK OK 13: 4.980 OK OK OK OK 14: 4.780 OK OK OK OK 15: 5.080 OK OK OK OK 16: 4.820 OK OK OK OK 17: 4.780 OK OK OK OK 18: 5.430 OK OK OK OK 19: 5.080 OK OK OK OK 20: 4.780 OK OK OK OK 21: 5.180 OK OK OK OK 22: 5.030 OK OK OK OK 23: 5.080 OK OK OK OK 24: 5.080 OK OK OK OK 25: 4.880 OK OK OK OK 26: 5.130 OK OK OK OK 27: 4.930 OK OK OK OK 28: 5.480 OK OK OK OK 29: 5.180 OK OK OK OK 30: 5.170 OK OK OK OK MAX: 5.480 MIN: 4.780 X bar: 5.09 S bar: 0.13 Pp: 2.5 Ppk: 1.72 OK / NG: 6000 / 6000 APPROVED: ISPRIANTI, CHANDRA	ITEM: Plating Thickness ALAT: X-Ray STD: Min 5.00, Max 5.00, MSK X L25, No Defect 1: 5.140 OK OK OK OK 2: 5.030 OK OK OK OK 3: 5.100 OK OK OK OK 4: 4.830 OK OK OK OK 5: 5.080 OK OK OK OK 6: 4.880 OK OK OK OK 7: 4.880 OK OK OK OK 8: 4.830 OK OK OK OK 9: 5.130 OK OK OK OK 10: 5.080 OK OK OK OK 11: 5.430 OK OK OK OK 12: 4.980 OK OK OK OK 13: 4.980 OK OK OK OK 14: 4.780 OK OK OK OK 15: 5.080 OK OK OK OK 16: 4.820 OK OK OK OK 17: 4.780 OK OK OK OK 18: 5.430 OK OK OK OK 19: 5.080 OK OK OK OK 20: 4.780 OK OK OK OK 21: 5.180 OK OK OK OK 22: 5.030 OK OK OK OK 23: 5.080 OK OK OK OK 24: 5.080 OK OK OK OK 25: 4.880 OK OK OK OK 26: 5.130 OK OK OK OK 27: 4.930 OK OK OK OK 28: 5.480 OK OK OK OK 29: 5.180 OK OK OK OK 30: 5.170 OK OK OK OK MAX: 5.480 MIN: 4.780 X bar: 5.09 S bar: 0.13 Pp: 2.66 Ppk: 1.83 OK / NG: 6000 / 6000 APPROVED: ISPRIANTI, CHANDRA

Conclusion:

All of Plating Thickness N30 Pp & Ppk for every matrix is OK

3. WWT Performance - Lab Scale (Mixing Tank WWT)

Number	Description	Lab Result				Internal Check (Pack Test)		
		Zn Content (ppm) Max 2 ppm	CN Content (ppm) Max 0.05 ppm	TDS (ppm)	TSS (ppm) Max 60 ppm	Zn Content (ppm) Max 2 ppm	CN Content (ppm) Max 0.05 ppm	Photos
Sample D	LC 34 (55gr/L) + Water After Filter Media + HCL 9,07% (from Acid Tank Line 1 & 2) + Ca(OH) ₂ + Polimer <u>Simulate Mixing Tank WWT</u>	0,338	<0,01 W/o NaClO Process	-	-	1 - 2	0.05 – 0.1 Without NaClO Process	
Sample G	Dasshi 39 (55gr/L) + Water After Filter Media + HCL 9,07% (from Acid Tank Line 1 & 2) + Ca(OH) ₂ + Polimer <u>Simulate Mixing Tank WWT</u>	0,177	<0,01 W/o NaClO Process	-	-	0 – 0.2	0.1 – 0.2 Without NaClO Process	
Sample J	PN 36 (55gr/L) + Water After Filter Media + HCL 9,07% (from Acid Tank Line 1 & 2) + Ca(OH) ₂ + Polimer <u>Simulate Mixing Tank WWT</u>	0,019	0,01 W/o NaClO Process	-	-	0 – 0.2	0.1 – 0.2 Without NaClO Process	

Conclusion:

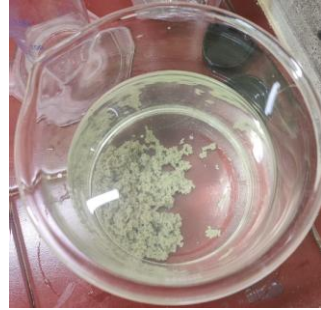
There is deviation of reading/measurement both in Zinc & Cyanide between pack test vs lab check

All chemicals Zinc content is OK both according to pack test & lab check result

All chemicals Cyanide contents is NG only according to pack test check. Notes : If we use NaClO like in our actual WWT possibly the Cyanide result will be much lower

LC-34 has the highest amount of Zinc compared to other chemicals, although still below the maximum government regulation

3. WWT Performance- Lab Scale (Flocculant Visual)

Number	Description	Flocculant Visual & Video
Sample D	LC 34 + Water After Filter Media + HCL 9,07% (from Acid Tank Line 1 & 2) + Ca(OH) ₂ + Polimer <u>Simulate Mixing Tank WWT</u>	  
Sample G	Dasshi 39 + Water After Filter Media + HCL 9,07% (from Acid Tank Line 1 & 2) + Ca(OH) ₂ + Polimer <u>Simulate Mixing Tank WWT</u>	
Sample J	PN 36 + Water After Filter Media + HCL 9,07% (from Acid Tank Line 1 & 2) + Ca(OH) ₂ + Polimer <u>Simulate Mixing Tank WWT</u>	

Conclusion:

All chemicals can build up flocculant during mixing tank WWT simulation. The result is OK

Based on the WWT performance result it is possible to proceed LC-34 without installing additional FeCl equipment in WWT

4. WWT Performance - Trial Masspro Scale



After trial LC-34 in plating, we store used LC-34 solution (79 gr/L) into separate drums.

We already get satisfactory result of WWT performance in lab scale

To ensure there is no discrepancy, in the future we will submit actual LC-34 which has been used in trial to independent lab in order to verify the result of previous WWT performance (lab scale)

Notes :

The process of the jar test, make purchase order, submission sample to independent lab & getting the analysis estimated around 2-3 weeks

4. WWT Performance - Trial Masspro Scale (Mixing Tank WWT)

Number	Description	Lab Result				Internal Check (Pack Test)		
		Zn Content Max 2 ppm	CN Content Max 0.05 ppm	TDS (ppm)	TSS (ppm) Max 60 ppm	Zn Content Max 2 ppm	CN Content Max 0.05 ppm	Photos
Sample O	LC 34 (69gr/L) + Water After Filter Media + HCL 9,07% (from Acid Tank Line 1 & 2) + Ca(OH) ₂ + Polimer <u>Simulate Mixing Tank WWT</u>	0,517	0,02 W/o NaClO Process	-	-	>5	>2 W/o NaClO Process	
Sample P3	LC 34 (100gr/L) + Water After Filter Media + HCL 9,07% (from Acid Tank Line 1 & 2) + Ca(OH) ₂ + H ₂ SO ₄ + NaClO + Polimer <u>Simulate Complete Mixing Tank WWT</u>	0,132	0,01	5183	14,2	2	<0.02	
Sample Q	LC 34 (69gr/L) + Water After Filter Media + HCL 9,07% (from Acid Tank Line 1 & 2) + Ca(OH) ₂ + H ₂ SO ₄ + NaClO + Polimer <u>Simulate Complete Mixing Tank WWT</u>	0,22	0,03	-	-	2 - 5	<0.02	

Conclusion:

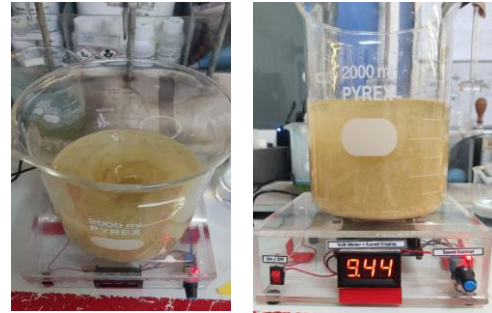
There is deviation of reading/measurement both in Zinc & Cyanide between pack test vs lab check

All samples Zinc content is OK according to lab check result

All samples Cyanide contents is OK according to lab check result

Notes : Our current pack test (both Zn & CN) already expired or near expired date which may affect the result except for the P3 sample

4. WWT Performance- Trial Masspro Scale (Flocculant Visual)

Number	Description	Flocculant Visual & Video
Sample O	LC 34 (69gr/L) + Water After Filter Media + HCL 9,07% (from Acid Tank Line 1 & 2) + Ca(OH) ₂ + Polimer <u>Simulate Mixing Tank WWT</u>	
Sample P3	LC 34 (100gr/L) + Water After Filter Media + HCL 9,07% (from Acid Tank Line 1 & 2) + Ca(OH) ₂ + H ₂ SO ₄ + NaClO + Polimer <u>Simulate Complete Mixing Tank WWT</u>	
Sample Q	LC 34 (69gr/L) + Water After Filter Media + HCL 9,07% (from Acid Tank Line 1 & 2) + Ca(OH) ₂ + H ₂ SO ₄ + NaClO + Polimer <u>Simulate Complete Mixing Tank WWT</u>	

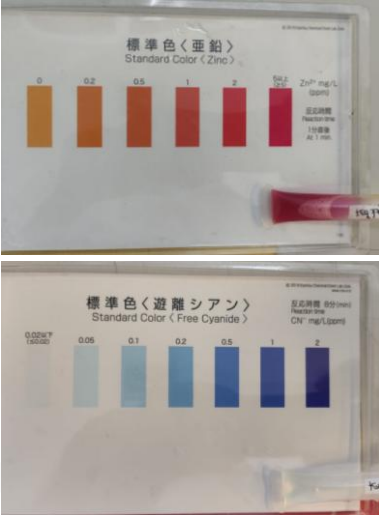
Conclusion:

All chemicals can build up flocculant during mixing tank WWT simulation. The result is OK

Based on the WWT performance result it is possible to proceed LC-34 without installing additional FeCl equipment in WWT

thank
you.

Comparison Pack Test Exp, New & Lab Test

Description	Lab Result		Internal Check (Pack Test Exp)			Internal Check (Pack Test New)		
	Zn Content Max 2 ppm	CN Content Max 0.05 ppm	Zn Content Max 2 ppm	CN Content Max 0.05 ppm	Photos	Zn Content Max 2 ppm	CN Content Max 0.05 ppm	Photos
Rinsing Water LC-34	0,03	0,01	>5	<0.02		>5	<0.02	

Conclusion:

- Using the same solution sample :
- Both Expired & New Pack Test (**Zinc**) resulting in almost identical color/value which way higher than Lab Result
 - Both Expired & New Pack Test (**Cyanide**) resulting in almost identical color/value and also almost similar with Lab Result
- The pack test is not accurate in detecting Zinc content of LC-34 but for detecting Cyanide the result is accurate**

Lampiran I : Peraturan Gubernur Provinsi Daerah Khusus
Ibukota Jakarta

Nomor 69 TAHUN 2013
Tanggal 16 Juli 2013

BAKU MUTU AIR LIMBAH BAGI KEGIATAN DAN/ATAU USAHA

A. Industri Pelapisan Logam

Parameter	Satuan	Kadar Maksimum	
		Pelapisan Logam	Galvanisasi
pH	-	6-9	6-9
TSS	mg/L	60	60
Cadmium (Cd)	mg/L	0,05	0,05
Sianida (CN)	mg/L	0,05	0,05
Krom Total (Cr)	mg/L	1	-
Krom Heksavalen (Cr ⁺⁶)	mg/L	0,1	-
Seng (Zn)	mg/L	2	2
Tembaga (Cu)	mg/L	1	1
Nikel (Ni)	mg/L	1	1
Zat Organik (KMnO ₄)	mg/L	50	50
COD	mg/L	75	75
Timbal (Pb)	mg/L	0,1	0,1
Perak (Ag)	mg/L	0,5	0,5